Inorganic Cumulative Exam

September 18, 2009

Questions posed by freshman taking chemistry often must be answered with qualifiers so that students are not misled. The qualifiers generally take the form of some exceptions to the general rule and the exceptions can often be used to reinforce the basic concepts. Answer the question below as if you were the instructor in general chemistry and provide examples of exceptions to your answer with some rationalization as why these are exceptions.

1. How can you tell whether a molecule will be ionically bonded or covalently bonded?

2. How can the periodic table be used to predict the electronic configuration of an atom?

3. What oxidation state do you expect for Group IA or 1 elements (H to Cs)?

4. What oxidation state do you expect for Group 18 elements (He-Xe)?